

CoRoT-2b

R-1158

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_220702 · created 2026-07-31T02:02:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459762.8424 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.065
Transit depth (Rp/Rs)^2	3.14 +/- 2.3 [%]
Orbital Inclination (inc)	89.6 +/- 2.04
Airmass coefficient 1 (a1)	0.026 +/- 0.015
Airmass coefficient 2 (a2)	2.98 +/- 1.03
Scatter in the residuals of the lightcurve fit is	58.57 %
Best Comparison Star	#1 - [569, 263]
Optimal Aperture	0.75
Optimal Annulus	4.75
Transit Duration (day)	0.0943 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.065 vs 0.1667 (deviation 0.15σ)
Duration fit vs published	0.0943 d vs 0.095 d (deviation 0.12σ)
Mid-time O–C	7.8 min
Depth SNR	2.6
Residual scatter	58.57 %

Light curve

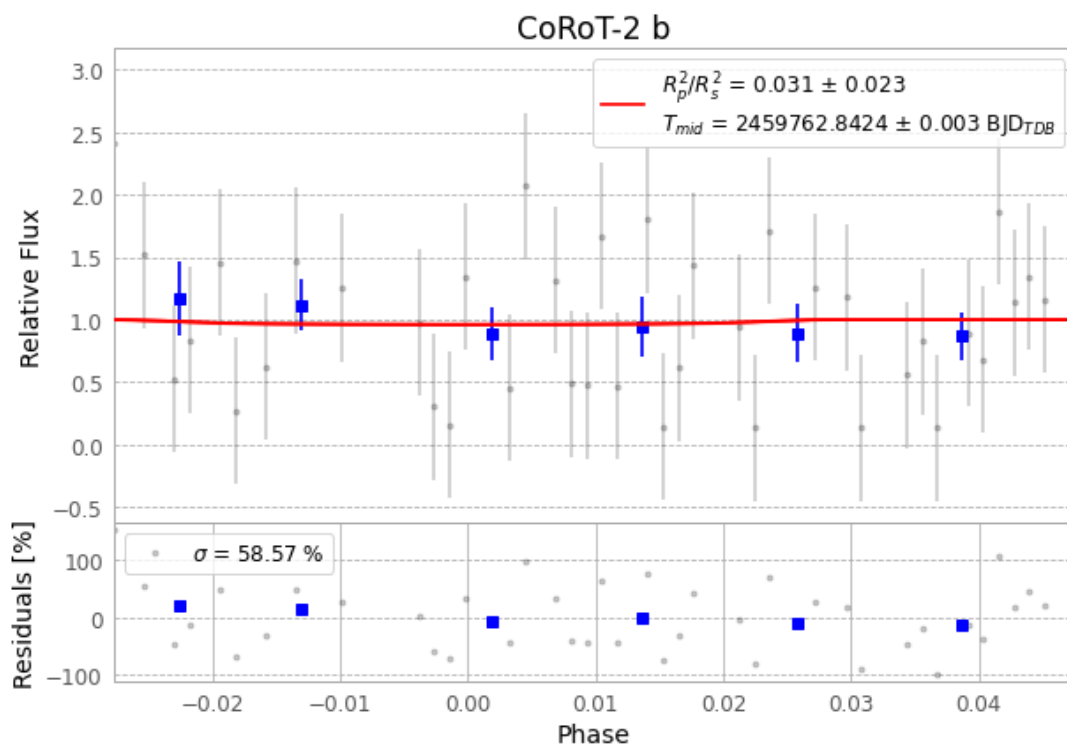


Figure 1 — FinalLightCurve_CoRoT-2 b_2022-07-01.png (SHA-256 a41238ba36fc9cfa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2fc0794f07c51431...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), CoRoT-2220702065413.FITS → CoRoT-2220702100613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-220702.log (7dc4e8f9e0eb...), FinalLightCurve_CoRoT-2 b_2022-07-01.png (a41238ba36fc...), FinalLightCurve_CoRoT-2 b_2022-07-01.csv (b5403886683d...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:02Z.